

Article Title

Ghost Infrastructure: Historical Industrial Geography and the Persistence of Path Dependent Accessibility in Bochum, Germany

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Acknowledgments

Not applicable.

Declaration of Generative AI and AI-Assisted Technologies

Generative AI (Claude, Anthropic) was used as an AI-assisted tool to inspect the GitHub repository for potential inconsistencies or contradictions between the documentation and the underlying code. The identified issues were subsequently reviewed and verified by the author, who independently addressed and resolved the bugs wherever possible. In cases where additional assistance was required, AI was used solely as a guidance and troubleshooting aid. Claude was also used to review and refine the manuscript for grammar, spelling, punctuation, and overall linguistic clarity.

Author Contributions

Sakshi D. Maske: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Writing – original draft; Writing – review & editing; Visualization; Project administration.

Statements and Declarations

Ethical considerations

Not applicable. This study did not involve human participants, human data, human tissue, or animal subjects. All data used are historical archival records (mine and worker-colony locations) and publicly available geospatial datasets (OpenStreetMap via OSMnx, GADM administrative boundaries). Ethics committee/IRB approval was therefore not required.

Consent to participate

Not applicable. No human participants were involved in this study.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data, images, or identifiable information.

Declaration of conflicting interest

I, the author have no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data availability

The data and code that support the findings of this study are openly available at https://github.com/sakshimaske303-commits/GHOST_INFRASTRUCTURE, with a permanently archived, version-controlled snapshot available via Zenodo at <https://doi.org/10.5281/zenodo.21761320>. This includes the historical GIS datasets (digitized coal mine and worker-colony locations for Bochum and Essen), the network accessibility and statistical analysis scripts, and all output data (JSON results files) referenced in this manuscript.

Author Biography

I am an independent geospatial researcher based in Maharashtra, India. I hold an undergraduate degree in Economics, Political Science, and History, and studied Geography as my optional subject while preparing for India's Civil Services Examination. Over the past year and a half, I have worked independently on applied geospatial research, using satellite data, GIS, spatial statistics, and Python-based computational methods to study questions in urban accessibility, environmental change, and historical land-use patterns. My projects are openly documented and reproducible, with an emphasis on transparent reporting of results, including null and partially replicating findings. PORTFOLIO : <https://geospatial-portfolio-three.vercel.app/>

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